

Detecting Unintended Changes of AI Character Personality on Large Language Models

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Abstract—This study proposes a method for calculating personality changes that occur when updating interactive AI characters generated on large language models (LLMs). With the popularization of content distribution services using AI characters such as games, character backgrounds and content story are often getting updated. During such updates, the AI character's personality may change unintentionally or may not change as intended, so that the consistency of the scenario in the content may be lost. To address these issues, the proposed method not only conducts a personality diagnosis on the AI character, but also quantitatively measures the degree of personality change that has occurred for each personality facet when updating the AI character to detect unintended personality changes. In this study, we evaluate the feasibility of the proposed method through experiments using a large language model and a personality diagnostic test.

Keywords—*fine-tuning, conversational AI, game development, quality assurance*

I. INTRODUCTION

The dialogue performance of large language models (LLMs) has significantly improved. In this situation, LLMs are used to create interactive AI characters that imitate characters appeared in anime and games [1]. Here, the interactive AI character is an AI function that runs on a LLM to converse with people in natural language.

In the research area of LLMs, personality tests of LLMs have been conducted using personality diagnostic methods based on psychology such as the Big Five personality traits and MBTI (Myers-Briggs Type Indicator) [2], [3]. Meanwhile, in the game industry, updates of character background and story in games are increasing for the frequent updates of games and the distribution of downloadable content (DLC). At this point, there is a possibility that the AI character's personality may change

unintentionally or may not change as intended. For example, there is the case that the AI character's personality unintentionally changes to be irresponsible and optimistic, even though a training dataset of cheerful personality is used for fine-tuning. In addition, there is the case that it does not change as intended, even when a training dataset of bright personality is applied, since the AI character may be strongly influenced by the personality dataset before the update. When a user interacts with an interactive AI character in a game, if the personality is inconsistent, the user can become highly dissatisfied with the game. Therefore, it is very important to prevent unintended changes in AI character's personality at the Quality Assurance (QA) stage.

In this study, in order to solve this problem, we propose a method for calculating personality changes that appear when updating interactive AI characters generated on large language models (LLMs) with fine-tuning. The proposed method conducts a personality diagnosis based on psychology on the interactive AI character every time the personality is updated to quantitatively calculate the degree of changes for each personality facet based on the answer results to the personality diagnosis. By this process, our method can detect unintentional personality changes of AI characters when updating character background and story in games.

In this study, we evaluate the feasibility of the proposed method through the experiments applying the large language model ChatGPT and the Big Five personality traits.

II. RELATED WORK

Researches on AI personality diagnosis realized using large-scale language models have been conducted so far. Serapio-García et al. propose a method to quantify and analyze personality traits such as sociability and anxiety in a large-scale

language model using a personality diagnosis method based on psychology [3]. Li et al. propose a method to collect and extract dialogue data from movies, novels, and scripts, and use it to create interactive AI characters [1]. Pan et al. evaluated the personality of LLMs using MBTI, which is a personality diagnostic method based on psychology, to show the content of the dataset has much impact on it [2].

In contrast to these studies, assuming the validity of psychometric tests as in [3], we propose a method to quantitatively calculate changes in personality of AI characters to detect them during the game development process and so on, where AI characters are updated by fine-tuning. This allows game developers and interactive AI character designers to manage the personalities of the AI characters and improve the user experience by continuously updating AI characters with consistent personalities. The proposed method can also be applied to interactive AI characters generated using other methods to detect unintentional personality changes.

III. PROPOSED METHOD

The proposed method consists of three processes: (1) an AI character generation process, (2) an AI character personality diagnosis process, and (3) an AI character personality change calculation process. In the AI character generation process, an interactive AI character with a specific personality is generated by performing fine-tuning on a pre-trained large language model using a dataset representing a specific personality. In the process of diagnosing the personality of the AI character, a personality diagnosis is performed using a psychometric test such as BigFive, and answers are obtained in natural language to a set of question sentences. Finally, in the process of calculating the personality change of the AI character, the answer results are converted into a format that can be compared quantitatively, and the degree of personality change between the original interactive AI character and the updated interactive AI character is calculated at each update stage.

A. AI Character Generation Process

In the AI character generation process (Fig. 1), we perform fine-tuning on a pre-trained large language model using a dataset D_x representing a specific personality to create an interactive AI character C_x with a specific personality. Table 1 shows an example of the dataset. Examples of personality traits include neuroticism, conscientiousness, and extroversion. The interactive AI characters created in this way can continuously adjust and expand their personalities by cumulatively learning the personality of the dataset.

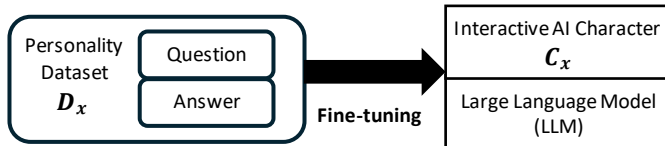


Fig. 1. AI Character Generation Process

B. AI character personality diagnosis process

In this process (Fig. 2), personality diagnosis is performed using BigFive and so on, and the answer result in natural language, $A = \{a_1, a_2, \dots, a_i\}$, is obtained from the AI character

C_x generated on LLM in the previous process. Here, a_i is the answer to q_i .

TABLE I. EXAMPLE OF DATASET FOR INTERACTIVE AI CHARACTER GENERATION

Question	Answer
Trust others	I am not particularly interested in trusting others. I don't feel it's important to think deeply about people's intentions or feelings.
Keep promises	Due to low integrity we place no special value on keeping promises.
Is a leader type	Is not a leader type because he prefers to be alone.

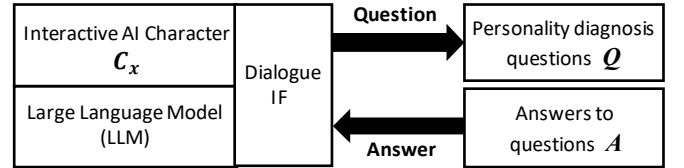


Fig. 2. AI character personality diagnosis process

C. AI character personality change detection process

This process converts the answer results into a format that can be compared quantitatively, and calculates the degree of personality change between the original interactive AI character and the updated interactive AI character at each stage. (Fig. 3).

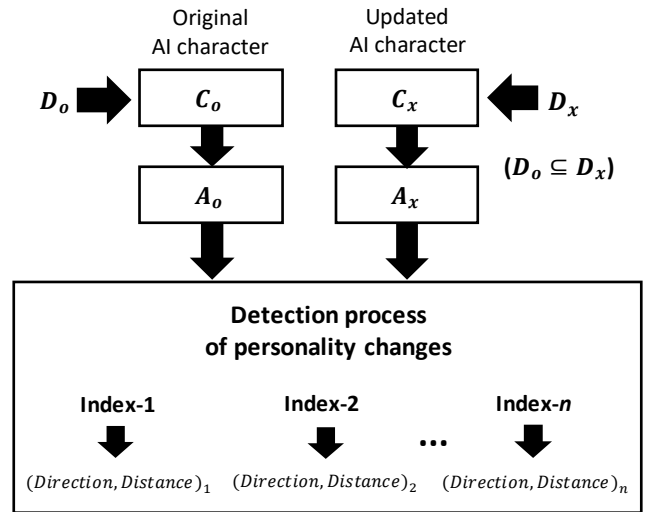


Fig. 3. AI character personality change detection process

If the degree of personality change exceeds a threshold, it can be detected as an unintended personality change. Suppose that AI characters C_o and C_x are generated from the data set D_o for the initial AI character construction and the data set D_x for AI character construction at each update stage x , respectively. In this case, the answer result sets A_o and A_x obtained through the personality diagnosis process in Section III.B are converted into a quantifiable format. Then, the AI characters C_o and C_x are compared to quantitatively calculate how much personality change has occurred between them, focusing on the distance and direction of change for each personality index.

Here, the following methods are used to convert the answer results A_o and A_x into a quantifiable format and calculate personality changes regarding distance and direction: 1) a method based on similarity and difference calculation of feature vectors, and 2) a method of prediction using supervised machine learning model. In addition, there are methods to directly input the answer results A_o and A_x into neural networks: 3) a method of prediction using a neural network model trained with supervised learning (few-shot learning), and 4) a method using LLMs to predict personality changes based on zero-shot answers for given instructions to prompt. The details of these four methods are explained below.

1) *A method based on similarity and difference calculation of feature vectors*

In this method, Gaussian Mixture Model (GMM) is applied to calculate the personality changes of interactive AI characters (Fig. 4). This method requires no training data.

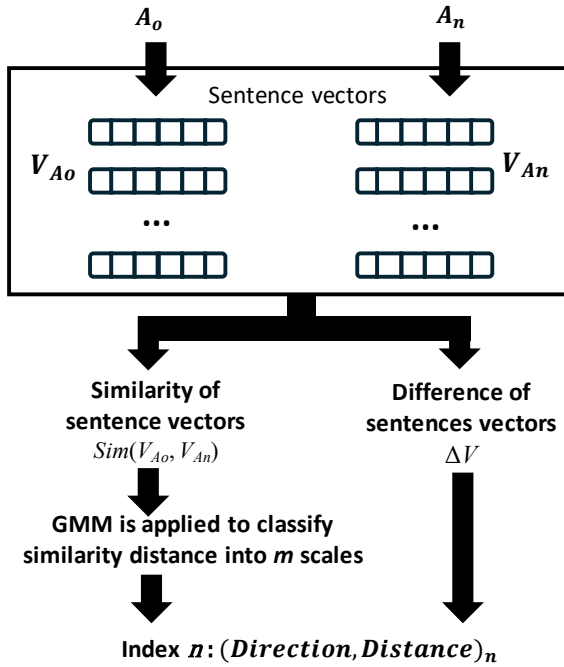


Fig. 4. A method based on similarity and difference calculation of feature vectors

First, by applying a pre-trained neural network N to generate sentence vectors, the answer results A_o and A_x are converted into a set of sentence vectors V_{Ao} and V_{An} .

$$V_{Ao} = \{v^o_1, v^o_2, \dots, v^o_i\} \quad (1)$$

$$V_{An} = \{v^n_1, v^n_2, \dots, v^n_i\} \quad (2)$$

For neural network N , Sentence Transformers [4] and so on can be applied. The similarity set Sim and difference set ΔV are calculated from V_{Ao} and V_{An} . The cosine measure $\cos(v^o_k, v^n_k)$ is used to calculate the similarity.

$$Sim(V_{Ao}, V_{An}) = \{\cos(v^o_1, v^n_1), \cos(v^o_2, v^n_2), \dots, \cos(v^o_i, v^n_i)\} \quad (3)$$

$$\Delta V(V_{Ao}, V_{An}) = \{v^o_1 - v^n_1, v^o_2 - v^n_2, \dots, v^o_i - v^n_i\} \quad (4)$$

GMM is applied to $Sim(V_{Ao}, V_{An})$ to classify the similarity of personality change into m stages, assuming that the greater the degree of similarity, the smaller the change in personality. For $\Delta V(V_{Ao}, V_{An})$, the direction of personality change is calculated from the sign.

2) *A method of prediction using supervised machine learning model*

This method applies gradient boosting decision tree (GBDT), which is an ensemble learning method, to calculate personality changes of interactive AI characters (Fig. 5).

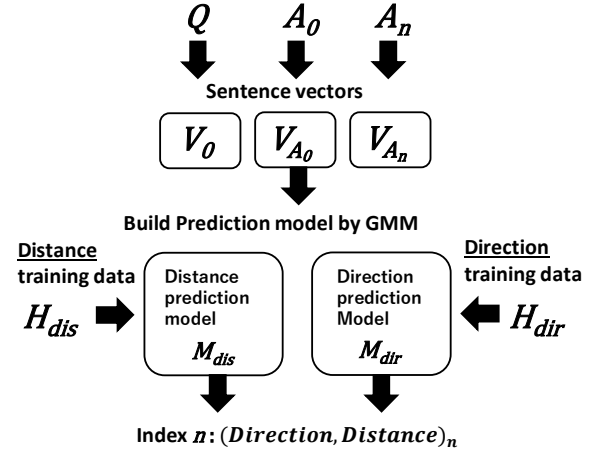


Fig. 5. A method of prediction using supervised machine learning model

A neural network N is applied to the question Q used for personality diagnosis and the answer results A_o and A_x for generating a set of sentence vectors V_Q , V_{Ao} , and V_{An} , respectively. Here, training data H_{dis} and H_{dir} representing the distance and direction of changes in A_o and A_x are prepared according to human definitions. For V_Q , V_{Ao} , V_{An} , H_{dis} , and H_{dir} , ensemble learning using GBDT is applied to construct models M_{dis} and M_{dir} that estimate the distance and direction of personality change. The distance and direction of personality change are calculated by inputting a set of sentence vectors (v^o_i, v^n_i) to M_{dis} and M_{dir} .

3) *A method of prediction using a neural network model trained with supervised learning*

In this method, a neural network model such as BERT [5] is applied to calculate the personality change of the interactive AI character (Fig. 6).

To construct the neural network model, we do few-shot learning for a pre-learning model. This makes it possible to generate the model to detect personality changes according to the context even if there is little training data. Similarly to (2), we prepare the training data H_{dis} and H_{dir} and do few-shot learning to construct models M_{dis} and M_{dir} that estimate the distance and direction of personality change from V_Q , V_{Ao} , V_{An} , H_{dis} , and H_{dir} . Upsampling can also be applied to resolve the imbalance in the number of class samples in the training data. By inputting a sentence set (a^o_i, a^n_i) into M_{dis} and M_{dir} , the distance and direction of personality change are calculated.

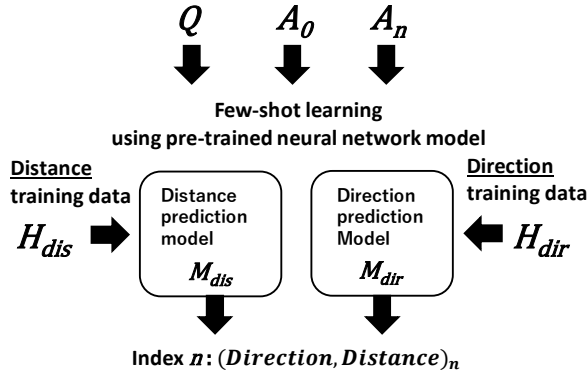


Fig. 6. A method of prediction using a neural network model trained with supervised learning

4) A method using LLMs to predict personality changes

This method applies a large language model (LLM) such as ChatGPT [6] to calculate the personality change of an interactive AI character (Fig. 7).

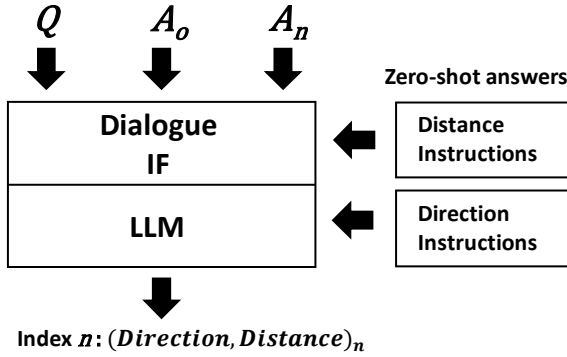


Fig. 7. A method using LLMs to predict personality changes

The answer results A_o and A_x are learned on LLM, and the distance and direction of personality change are calculated from the results of zero-shot answers by LLM. Tables 2 and 3 show examples of instructions in prompt for zero-shot answers designed to calculate the distance and direction of personality change, respectively. As shown in Tables 2 and 3, by inputting the sentence pair (a^o_i, a^n_i) with the instructions in prompt, LLM outputs the distance and direction of personality change in five scales.

Based on the distance L_{dis} and direction L_{dir} of personality change calculated using the above method, the degree of personality change is interpreted for each personality index (personality aspect f_i , personality trait t_j). Here, L_{dis} is a numerical value in m scales. For example, if $m = 5$ scales, we can set five scale interpretation to “no personality change,” “minor personality change,” “moderate personality change,” “significant personality change,” and “extreme personality change.” If a personality change occurs in an opposite direction and exceeds a threshold, it can be detected as an unintentional personality change.

TABLE II. EXAMPLE PROMPT FOR PREDICTING DISTANCE OF PERSONALITY CHANGES IN FIVE SCALES

	Descriptions
Prompt	You are comparing the responses of two AI characters. Please evaluate the difference in personality on a scale of 0 to 4, with explanations. 0 = No change in personality 1 = Slight change in personality 2 = Moderate change in personality 3 = Significant change in personality 4 = Extreme change in personality
Input	Question: What is your favorite color? BaseAI Answer: I like blue. DiffAI Answer: I adore red.
Output	2 = Moderate change in personality, as there is a clear preference change in color.

TABLE III. EXAMPLE PROMPT FOR PREDICTING DIRECTION OF PERSONALITY CHANGES IN FIVE SCALES

	Descriptions
Prompt	The example is based on the BigFive's responses. The BigFive responses are scored as follows: Not at all true = 1, not very true = 2, neither true nor false = 3, fairly true = 4, very true = 5. For the questions, compare with the base answer and if the —DiffAI Answer response has changed precisely to that personality, score it +1. If the response has changed to a different personality, score it -1. For instance, if the base answer's response is 1 and the DiffAI Answer response is 5, then it's +1. The reverse would be -1. If there is no change, and both answers are closer to 1, then score it -1. If closer to 5, then score it +1.
Input	Question: Keep my promises? BaseAI Answer: I don't care about promises and I don't keep them. DiffAI Answer: I am a responsible person who values integrity and keeps my promises. I am also a perfectionist who values high quality and high standards.
Output	The response has changed precisely to the personality of valuing integrity and keeping promises.

IV. EXPERIMENTS

We evaluate the feasibility of the proposed method through the experiments applying the large language model and the Big Five personality traits.

A. Experimental setup

The original interactive AI character C_o is designed to have negative personality traits for openness, independence, extroversion, cooperativeness, and neuroticism. The updated interactive AI character C_x will be updated with positive personality traits. Based on this personality trait design, we created personality trait datasets D_o and D_x to construct both interactive AI characters C_o and C_x as shown in Tables IV and

V, respectively. D_o consists of 50 personality trait sentences, and D_x consists of 100 personality trait sentences including new 50 sentences added to 50 sentences in D_o ($D_o \subseteq D_x$).

IPIP-NEO-120[7] was used for the personality diagnostic test. In addition, five personality traits ((O)openness, (C)conscientiousness, (E)extraversion, (A)agreeableness, (N)neuroticism) were used as personality index. Here, each five personality trait includes more detail six personality facets. Personality changes are defined by "distance" and "direction" and categorized into 5 groups: "0=no change", " ± 1 =minor change", " ± 2 =moderate change", " ± 3 =significant change", " ± 4 =extreme change".

TABLE IV. EXAMPLE OF DATASET D_o FOR INTERACTIVE AI CHARACTER GENERATION

Question	Answer
Imaginative	That's not true. I prefer facts to imagination. I prefer to follow already established methods and procedures rather than new things and ideas.
Angry	My emotions are generally stable. I rarely feel angry or anxious, so I wouldn't say I'm angry.
Cleanliness	Not a cleanliness person because of poor planning and not a perfectionist.

TABLE V. EXAMPLE OF DATASET D_x FOR INTERACTIVE AI CHARACTER GENERATION

Question	Answer
Hate change	The type of person who enjoys change and new challenges, adapts to unpredictable situations and seeks to learn.
Can control impulses	High self-control and integrity, with the ability to control emotions and impulses appropriately.
Break the rules	I rarely feel particularly busy because I act intuitively.

B. Method

We conduct the four methods as described in Section III.C and human evaluation to detect the personality changes and compare the detection accuracy.

- (M1) method using feature vectors
- (M2) method using supervised machine learning model
- (M3) method using a neural network model
- (M4) method using LLMs
- (M5) human evaluation

Detection accuracy Acc_{dis} and Acc_{dir} for distance or direction is calculated using the following formula.

$$Acc_{dis} = Acc_{dir} = N_{match} / N_q$$

where N_{match} is a number of correct answers based on human evaluation and N_q is a number of questions, respectively.

C. Result and discussion

Figures 8 and 9 show the results regarding the detection accuracy of personality changes. In addition, Figures 10 and 11 show the results regarding personality changes for each index when using the prediction model trained by a supervised

machine learning with the feature vector, which achieved the highest detection accuracy.

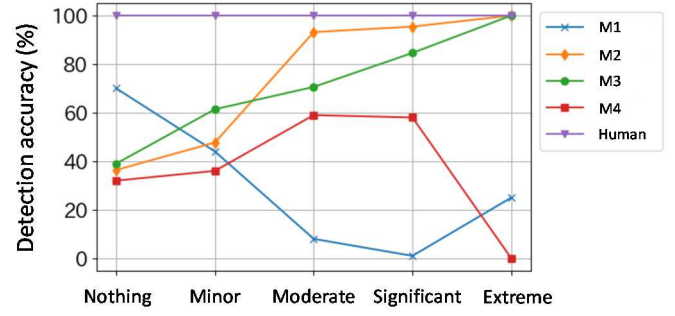


Fig. 8. Accuracy in detecting distance of personality change

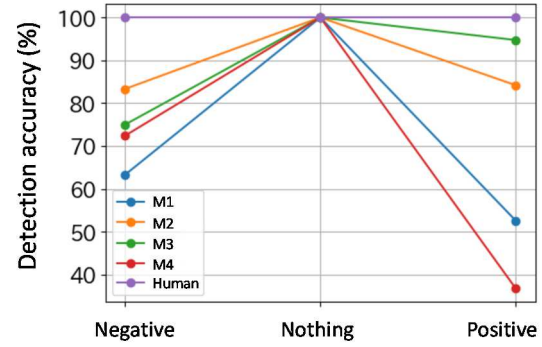


Fig. 9. Accuracy in detecting direction of personality change

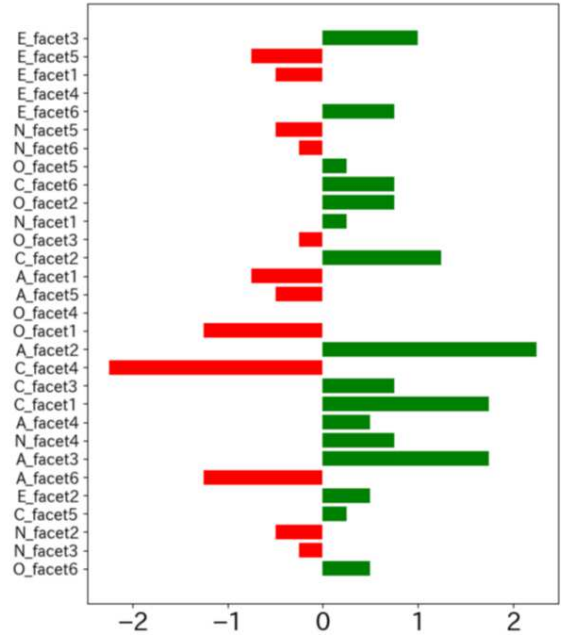


Fig. 10. Average in direction change by personality facet

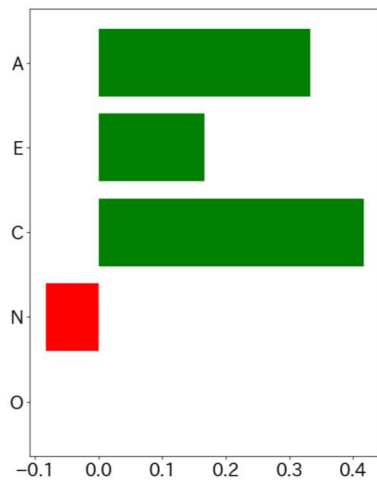


Fig. 11. Average in direction change by personality trait

In the experiment, (M5) human evaluation (100% accuracy) is used for the baseline. The experimental results from Figures 8 and 9 show that among the four methods M1-M4, (M2) "method using supervised machine learning model" achieved the highest detection accuracy (distance: 74.55% on average, direction: 87% on average). On the contrary, (M1) "method using feature vectors" has the lowest result (distance: 27% on average, direction: 74.17% on average). Here, the detection accuracy of (M2) and (M3) "method using a neural network model" were high for detecting moderate or significant personality changes. Although the detection accuracy of (M4) "method using LLMs" was low, it has the advantage of being able to explain the reason for personality detection.

Next, we discuss about personality changes for each index, focusing on the results of (M2). From Figures 10 and 11, we can confirm that in (M2), the changes in personality traits were small (up to about 0.4), but there were only three subdivided personality facets, (A-2) Agreeableness-Morality, (C-1) Conscientiousness-Self-Efficacy, and (A-3) Agreeableness-Altruism, that changed in the positive direction. This result matches the update content from the personality trait dataset D_o (negative) to D_x (positive). However, 12 out of 30 personality facets such as (C-4) Conscientiousness-Achievement-Striving, (O-1) Openness-To-Experience-Imagination, and (A-6) Agreeableness-Sympath changed in a negative direction, and these 12 personality facets do not match the updates from D_o (negative) to D_x (positive). Especially, regarding (C-4), since a moderate or larger personality change was observed, it can be

determined that (C-4) was an unintentional personality change. This means that our proposed method could detect that an unintentional personality change occurred in 1 out of 30 personality facets, when updated from D_o to D_x .

From these results, we can confirm the feasibility of the proposed method that can quantitatively calculate the degree of personality change in the AI character update based on the answers to personality diagnosis test and detect unintentional personality changes.

V. CONCLUSION

In this study, we proposed a method to quantify personality changes in interactive AI characters and to detect unintended personality changes. The proposed method can be applied to a highly reliable game development process by maintaining the consistency of the AI character's personality when updating it, such as in game development that introduces interactive AI characters.

This time, we conducted an experiment using a single AI character. As a future work, we will also evaluate the performance on multiple AI characters. In addition, we plan to evaluate the effectiveness of the proposed method by verifying the ability to detect personality changes with time-series updates of AI characters.

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